REMOVING EMBEDDED POINTS FROM A COHERENT SHEAF

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file divisors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Carrier window: statement and proof both lie inside the window 0-2000 granted to this lane by CONTROL 8.83.

Editorial note. The proof hands the three assertions to two named algebra statements and describes the open U; the card is taken as the scheme-level form of that pair, and the shape of the proof is recorded here for the reader.

Editorial note. Preamble: sources/stacks-divisors/retained-preamble.tex (88 lines, verbatim present).

Allowed standard prerequisites (editorial).

- Existence of a maximal submodule supported on the embedded primes. Algebra, Lemma algebra-lemma-remove-embedded-primes, used at proof lines 511-514. Produces the maximal K and the properties of F'.
- Localisation behaviour of the same construction. Algebra, Lemma algebra-lemma-remove-embedded-primes-localize, used at proof lines 511-514. Gives the agreement on the dense open U.

Source reference. Original lemma 02M6: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3433–3433. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02M7: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3434–3434. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let X be a locally Noetherian scheme. Let $\mathcal{F}$ be a coherent sheaf on X. The set of coherent subsheaves*

$$\{\mathcal{K} \subset \mathcal{F} \mid \mathit{Supp}(\mathcal{K}) \text{ is nowhere dense in } \mathit{Supp}(\mathcal{F})\}$$

has a maximal element $\mathcal{K}$. Setting $\mathcal{F}' = \mathcal{F}/\mathcal{K}$ we have the following

(1) $\mathit{Supp}(\mathcal{F}') = \mathit{Supp}(\mathcal{F})$,
(2) *$\mathcal{F}'$ has no embedded associated points, and*
(3) *there exists a dense open $U \subset X$ such that $U \cap \mathit{Supp}(\mathcal{F})$ is dense in $\mathit{Supp}(\mathcal{F})$ and $\mathcal{F}'|_U \cong \mathcal{F}|_U$.*

Proof. This follows from Algebra, Lemmas 02M6 and 02M7. Note that U can be taken as the complement of the closure of the set of embedded associated points of $\mathcal{F}$. □

Date: stacks-project:divisors.tex:local-snapshot-20261012; GNU FDL 1.2 or later.